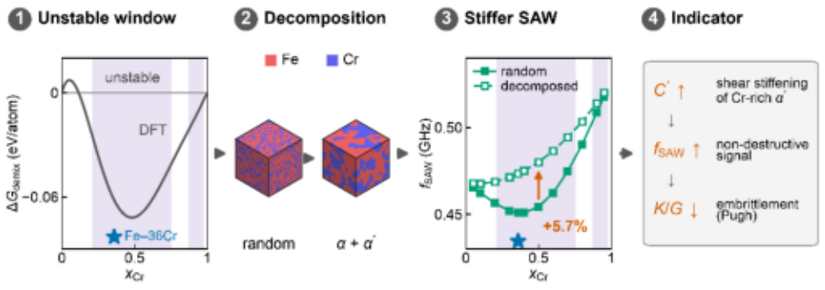
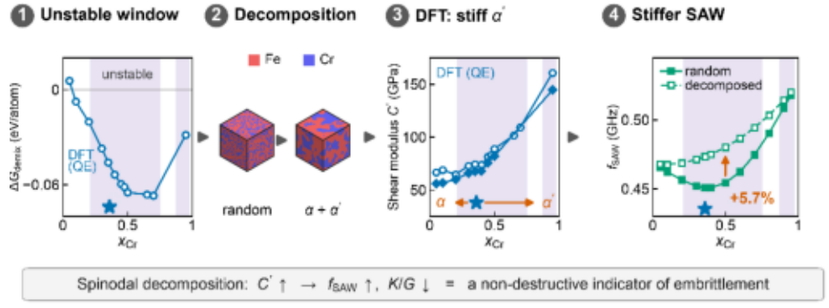
Graphical abstract: all versions (newest last)

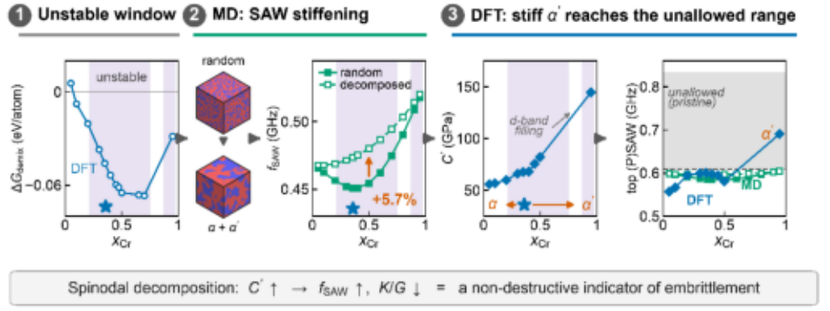
v1



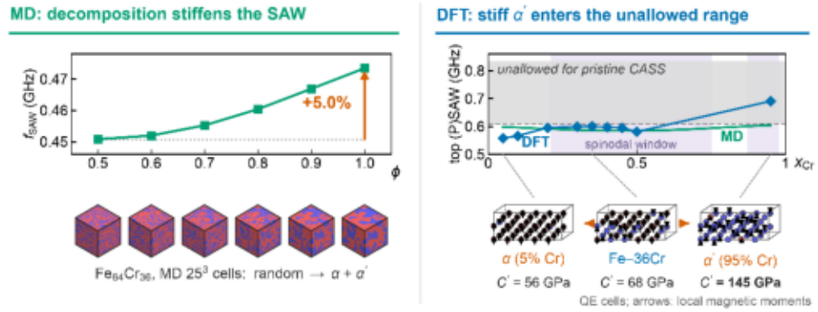
v2



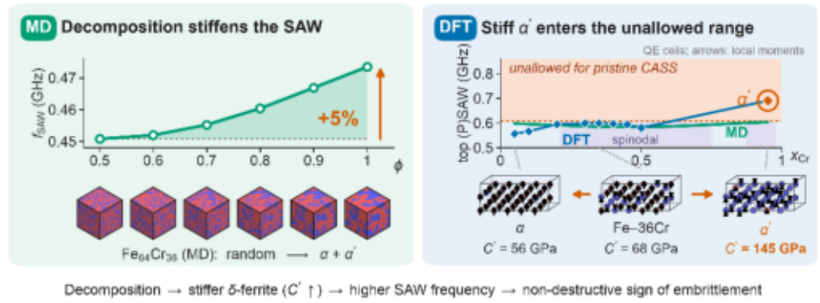
v3



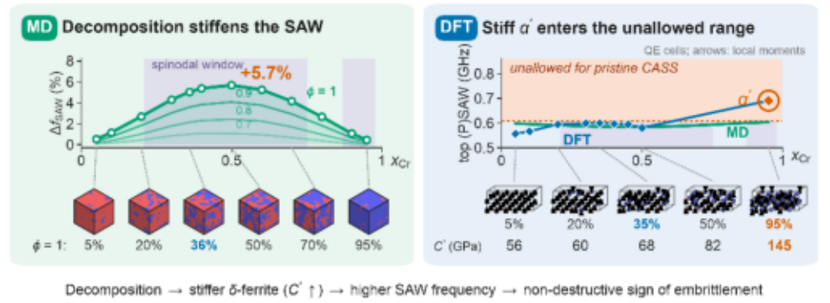
v4



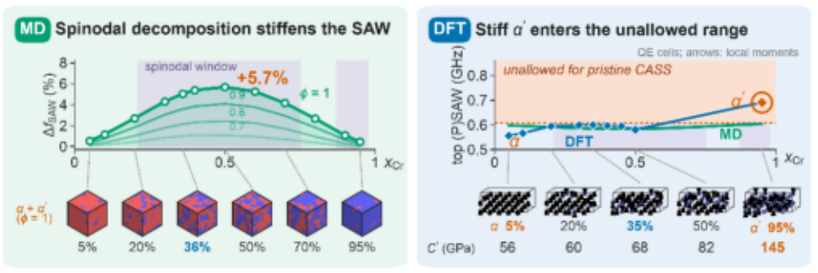
v5



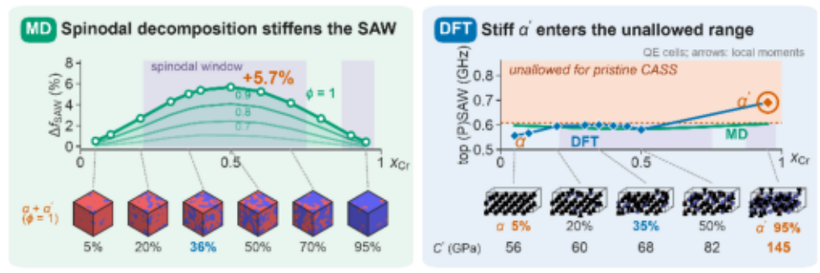
v6



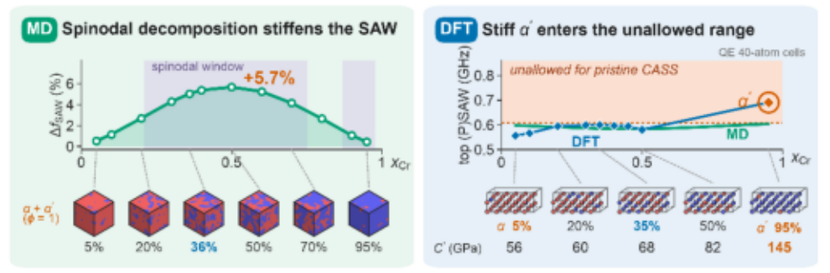
v7



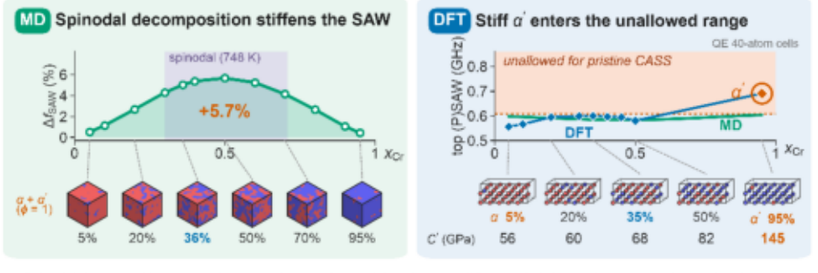
v8



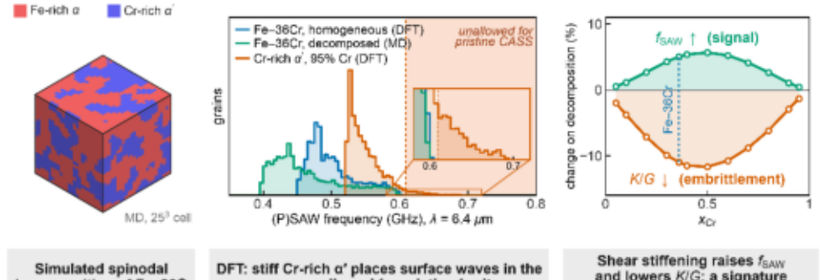
v9



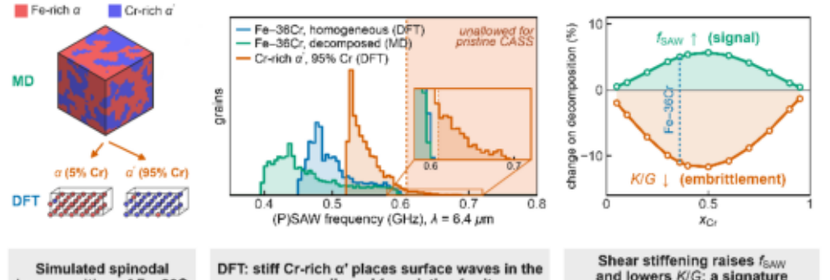
v10



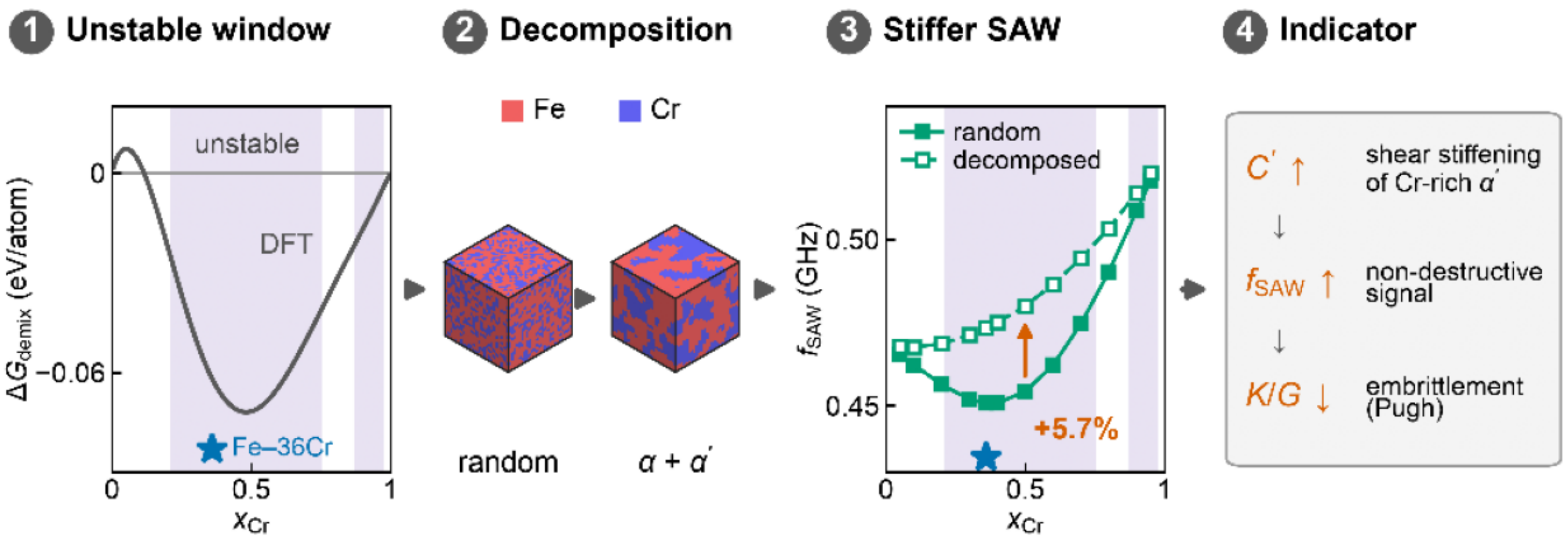
v11



v12

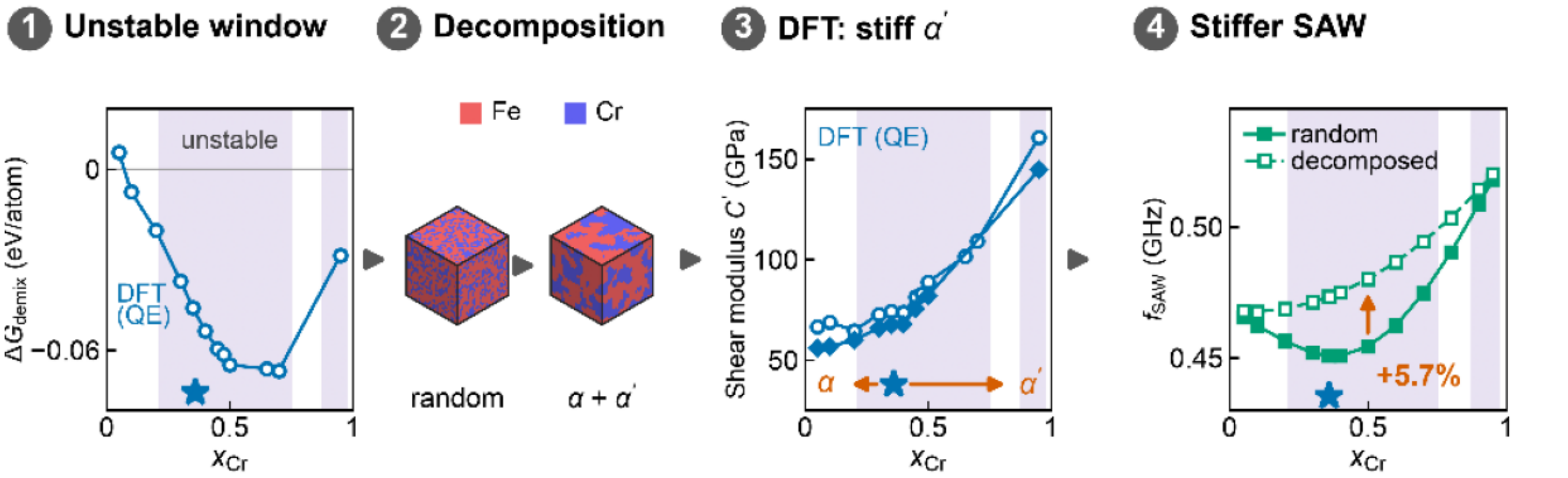


v1



First draft: four numbered steps (unstable window, MD cubes, SAW stiffening, indicator box)

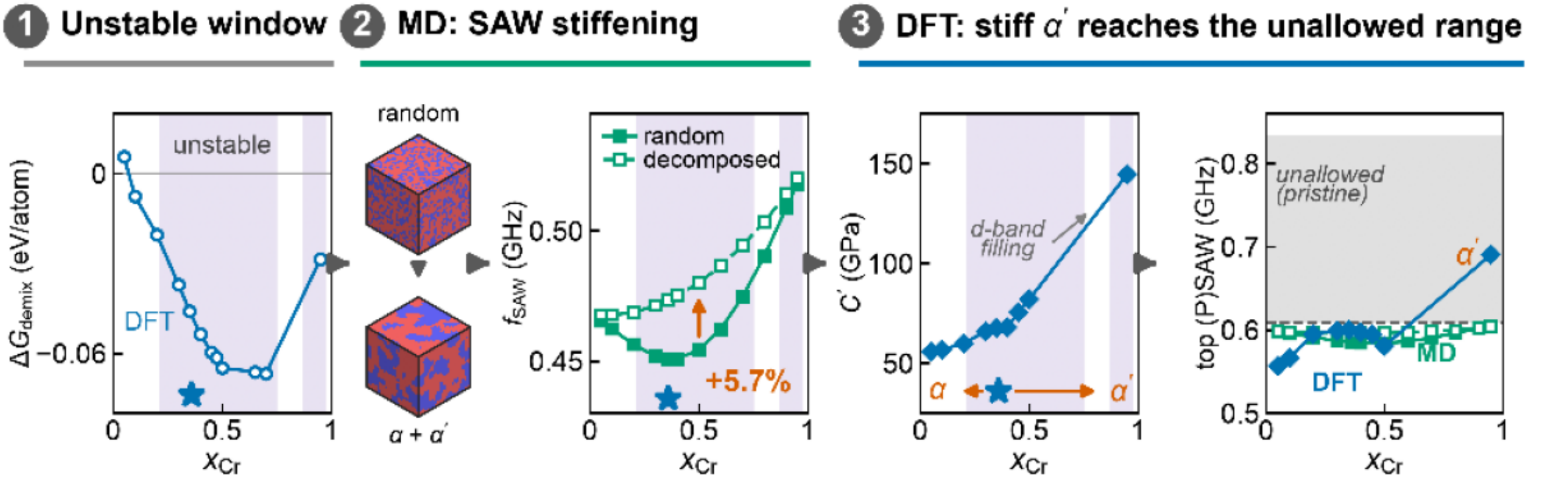
v2



Spinodal decomposition: $C' \uparrow \rightarrow f_{SAW} \uparrow$, $K/G \downarrow$ = a non-destructive indicator of embrittlement

DFT added: QE demixing energy (1) and QE C' (3); conclusion as banner

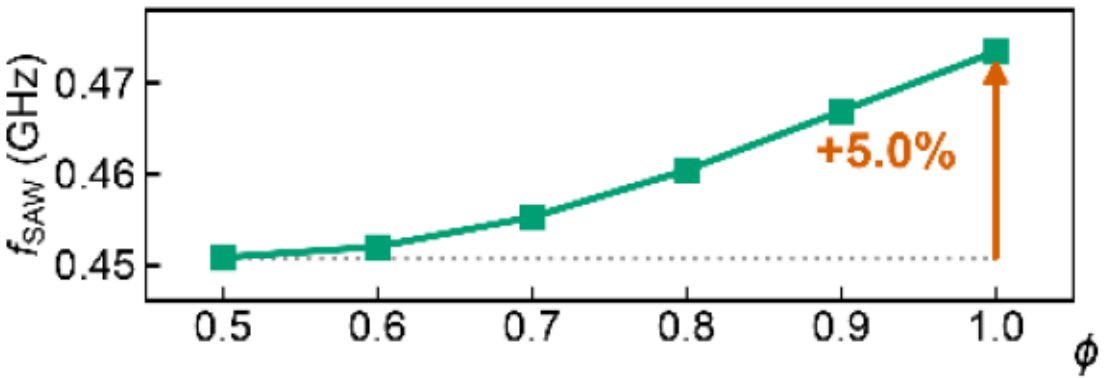
v3

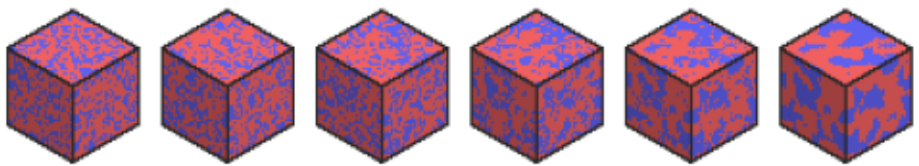


MD and DFT side by side; DFT top (P)SAW entering the unallowed range (Fig. 2b)

v4

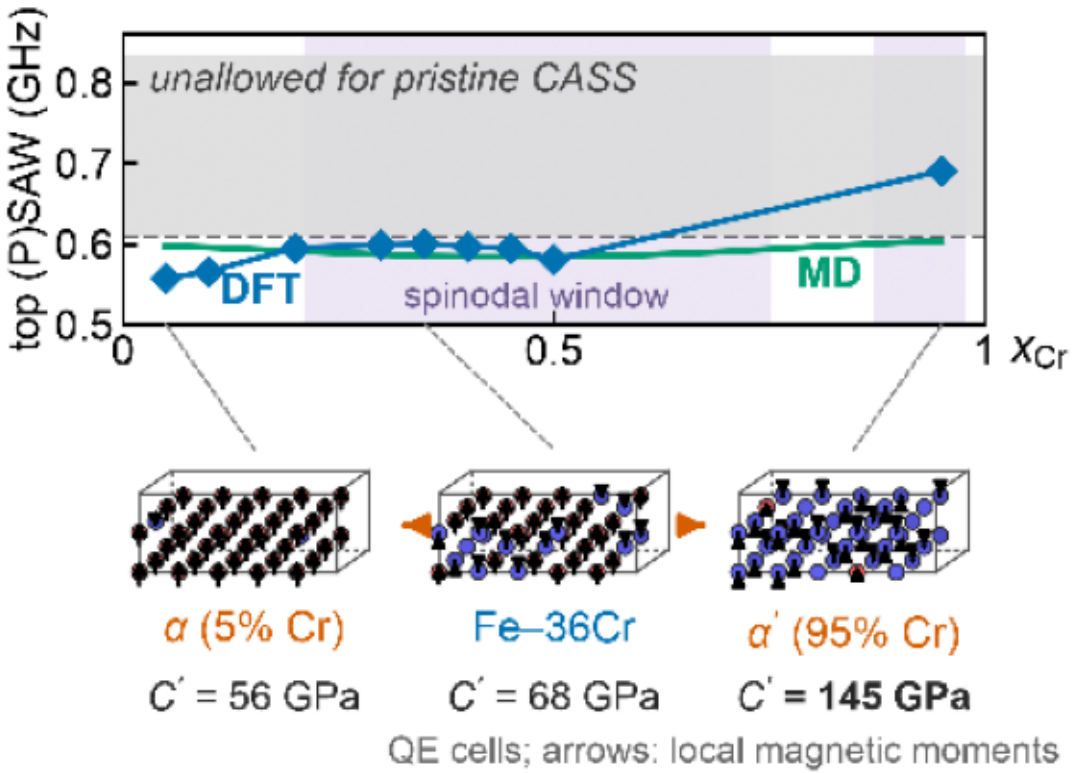
MD: decomposition stiffens the SAW





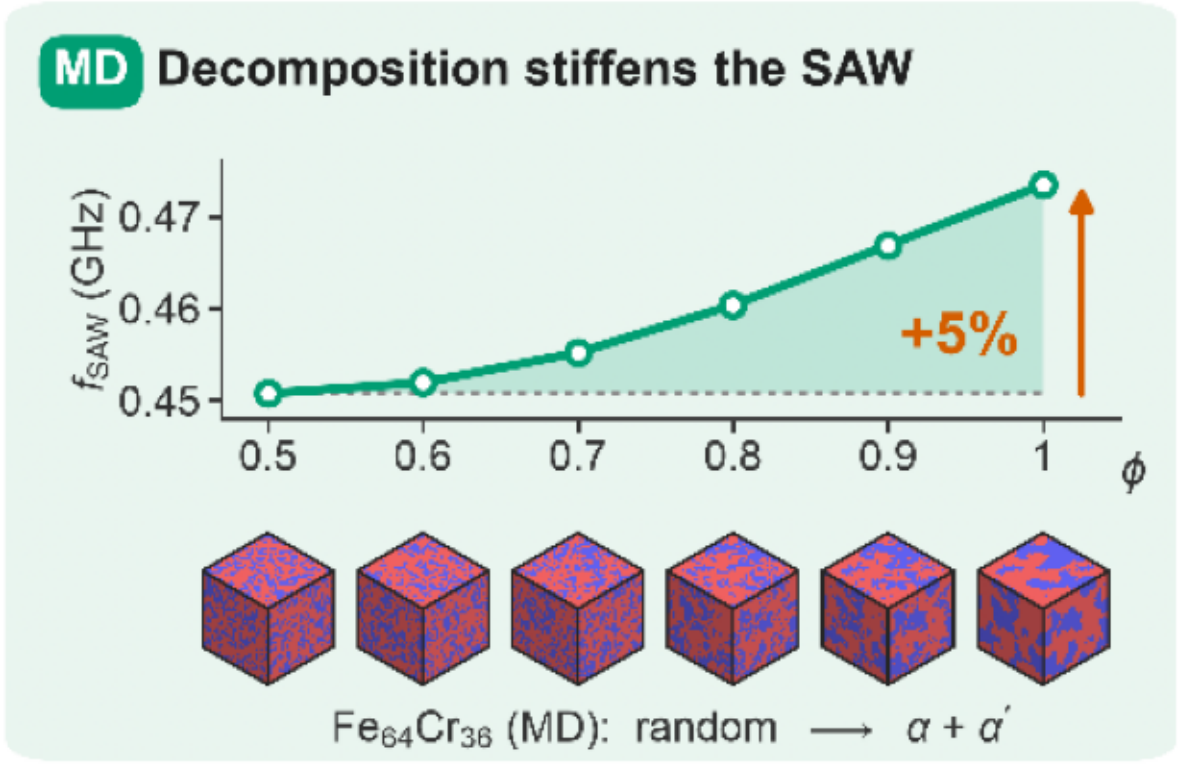
$Fe_{64}Cr_{36}$, MD 25^3 cells: random → $\alpha + \alpha'$

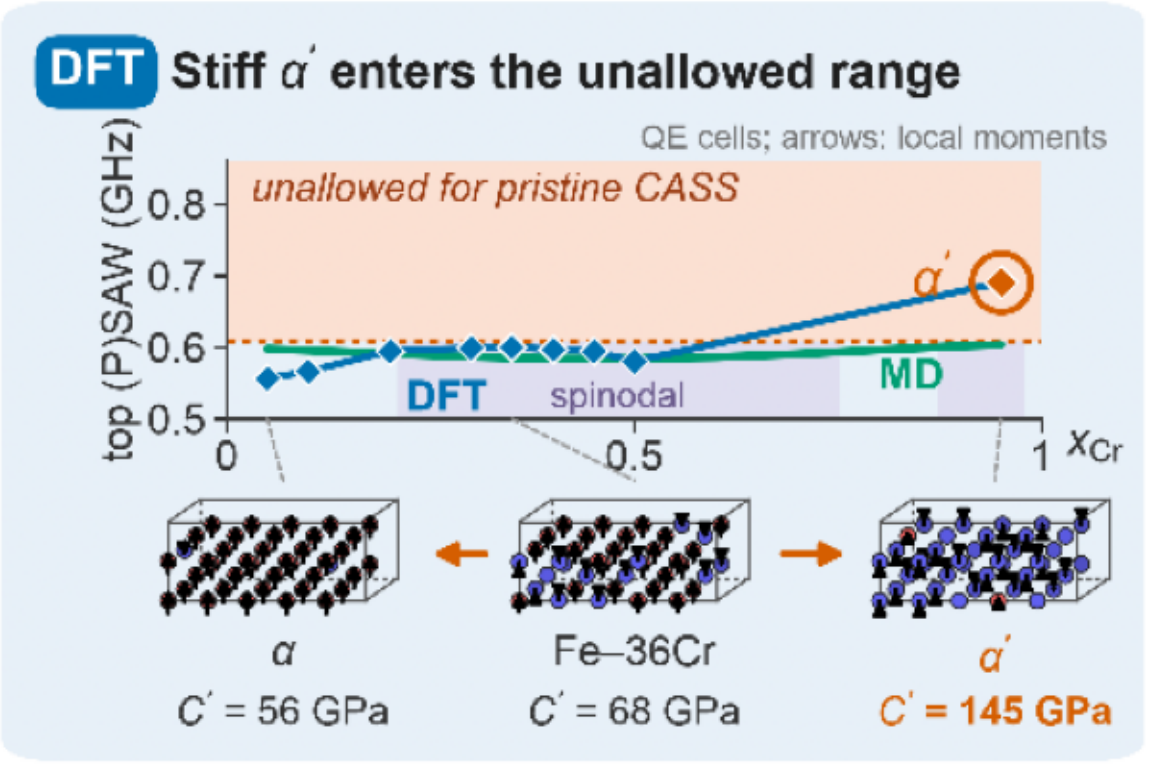
DFT: stiff α' enters the unallowed range



Two halves: key result plotted above its structures (MD cubes along phi; QE cells alpha, Fe-36Cr, alpha')

v5

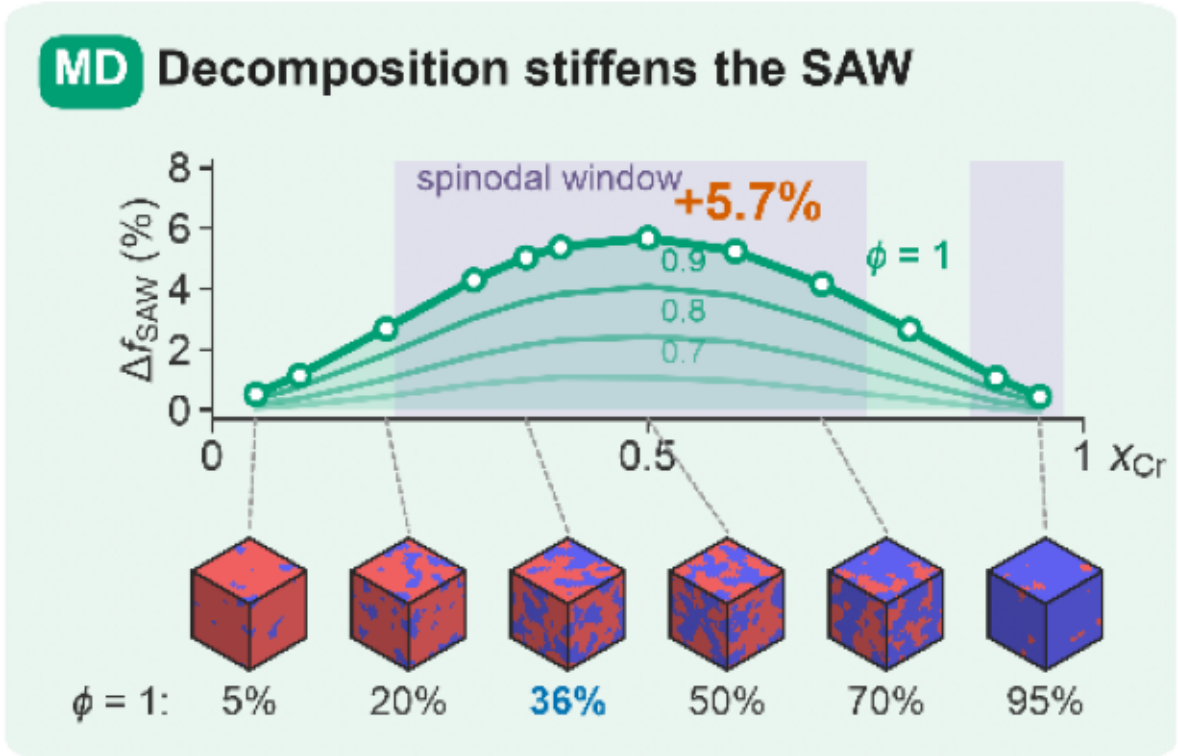


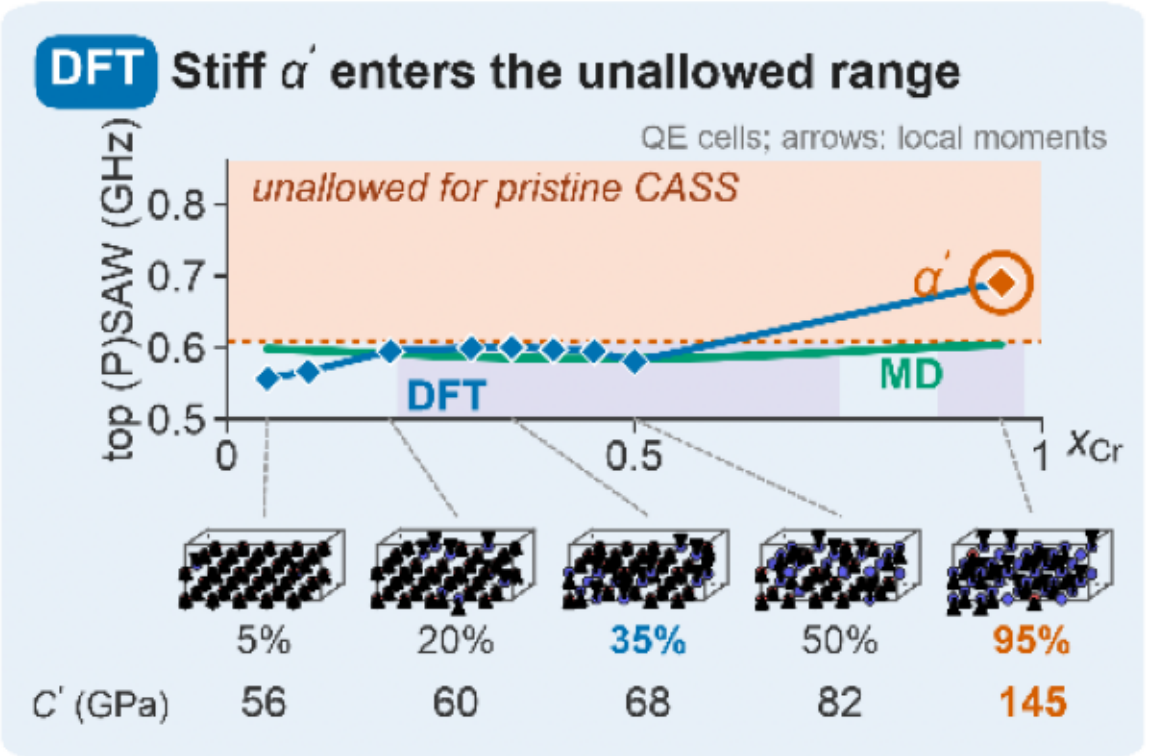


Decomposition → stiffer δ-ferrite (C' ↑) → higher SAW frequency → non-destructive sign of embrittlement

v4 restyled: tinted cards, MD/DFT tags, open axes, highlighted alpha', banner

v6

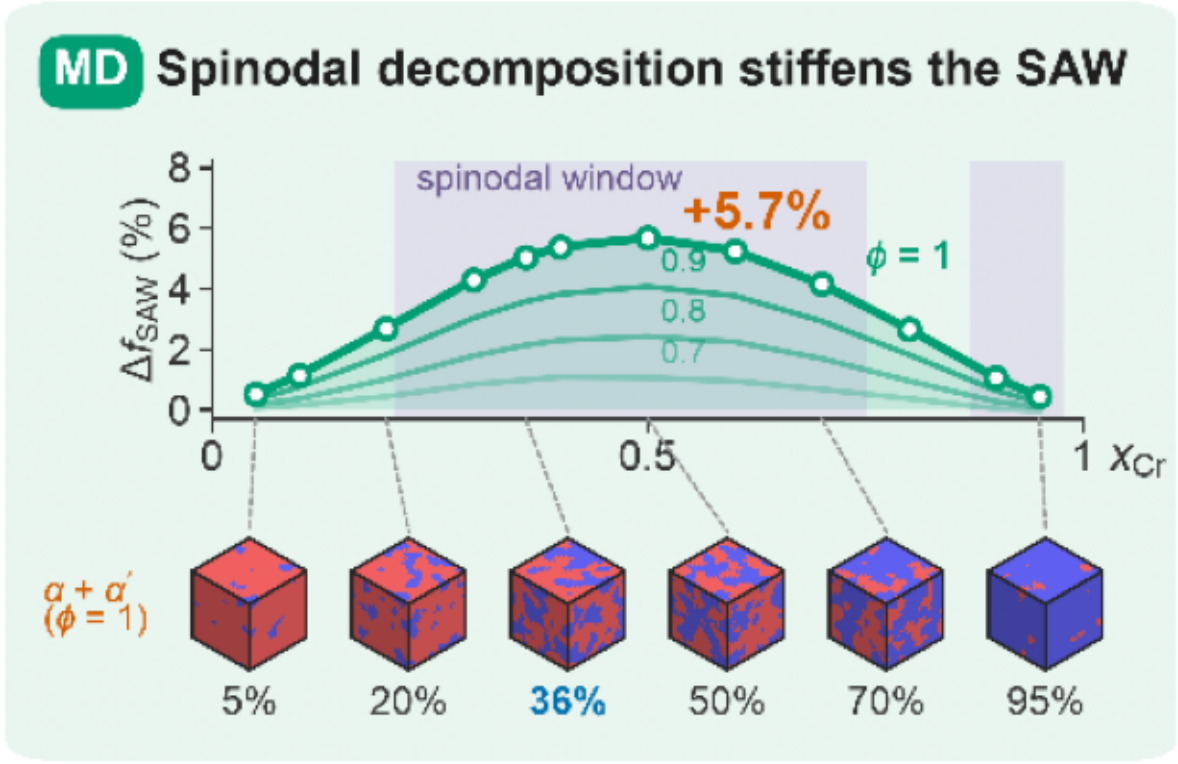


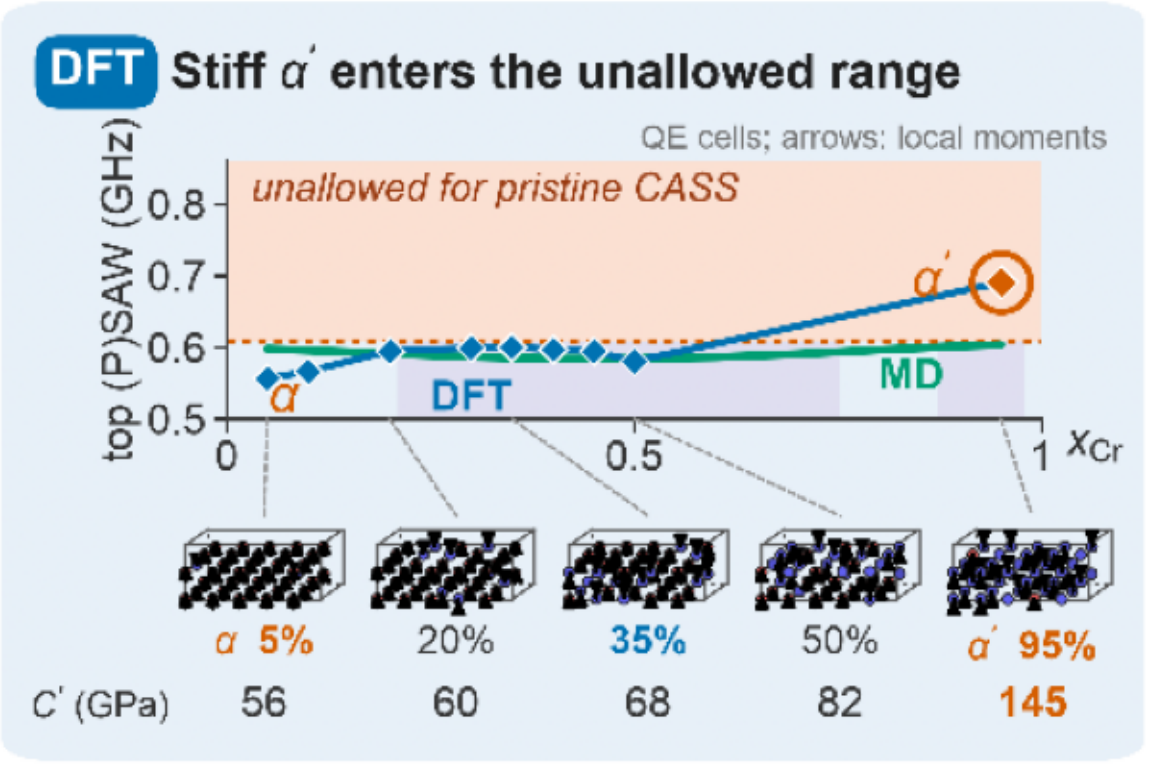


Decomposition → stiffer δ-ferrite (C' ↑) → higher SAW frequency → non-destructive sign of embrittlement

Both cards on one x_Cr axis: all twelve MD compositions (phi = 0.7-1), six MD cells, five QE cells with C'

v7

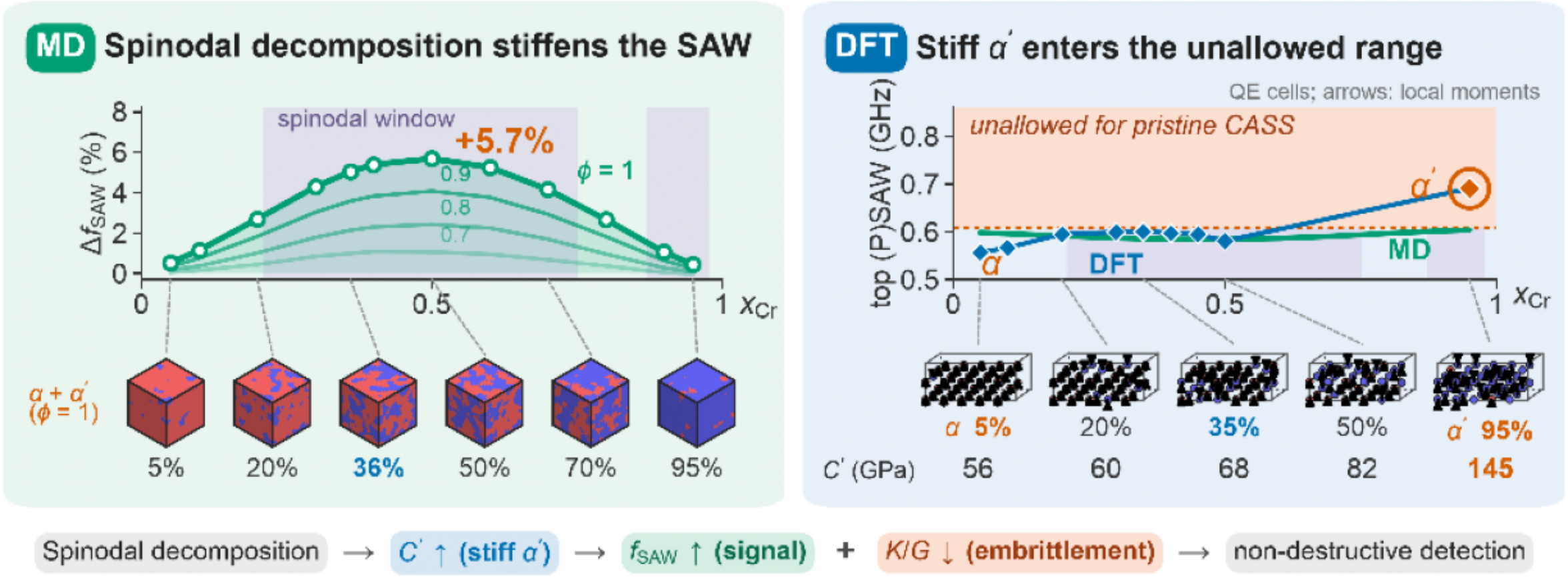




Spinodal decomposition → stiffer δ-ferrite ($C' \uparrow$) → higher SAW frequency → non-destructive sign of embrittlement

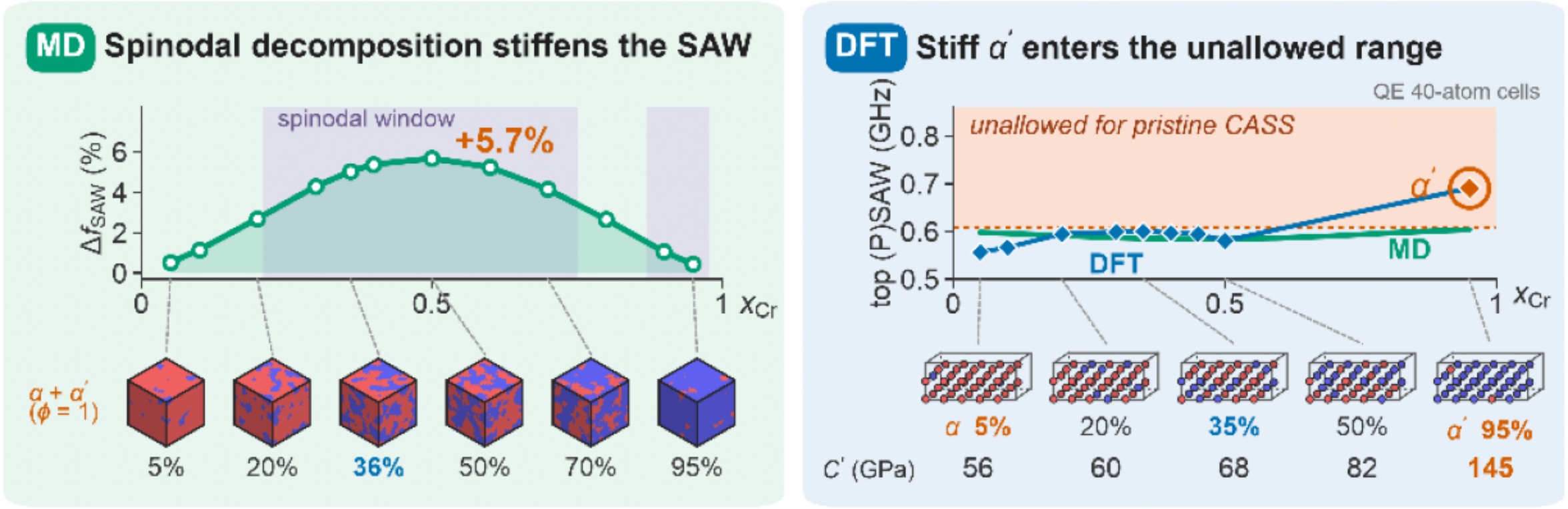
"Spinodal decomposition" in title and banner; alpha / alpha' labels back

v8



Mechanism line as coloured chips including K/G down (embrittlement)

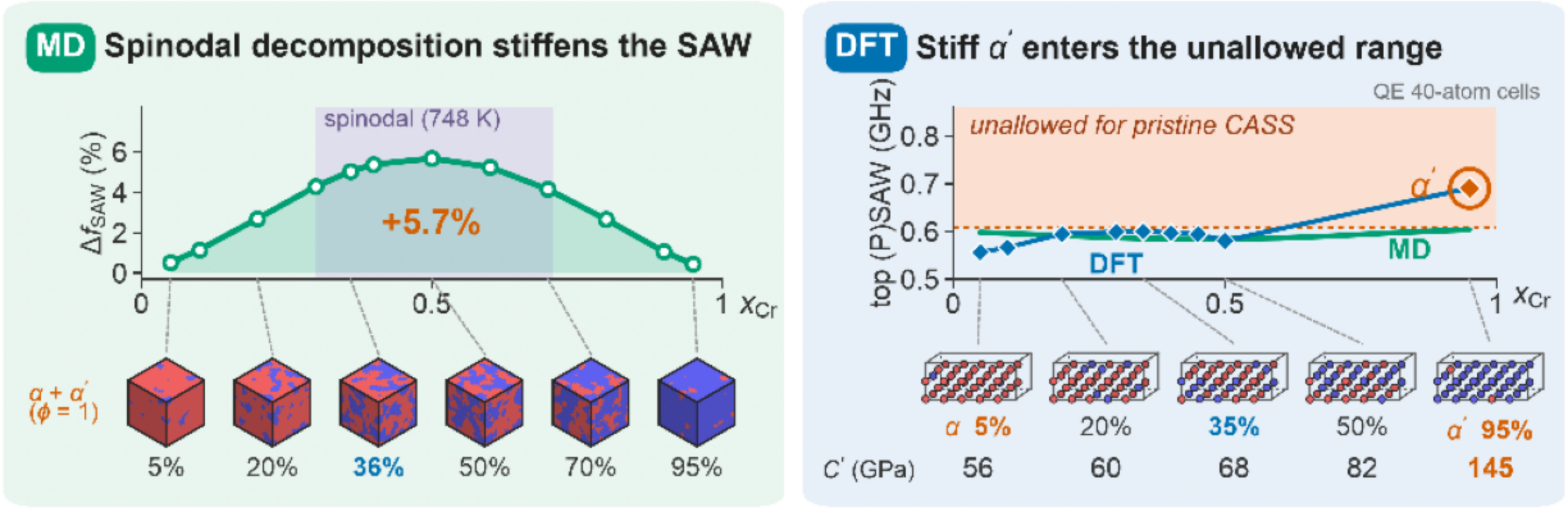
v9



Spinodal decomposition → C' ↑ **(stiff α')** → f_{SAW} ↑ **(signal)** + K/G ↓ **(embrittlement)** → non-destructive detection

Decluttered v8: phi = 1 curve only, QE cells without moment arrows, no window shading or extra alpha label in the DFT plot, mechanism line as plain coloured text

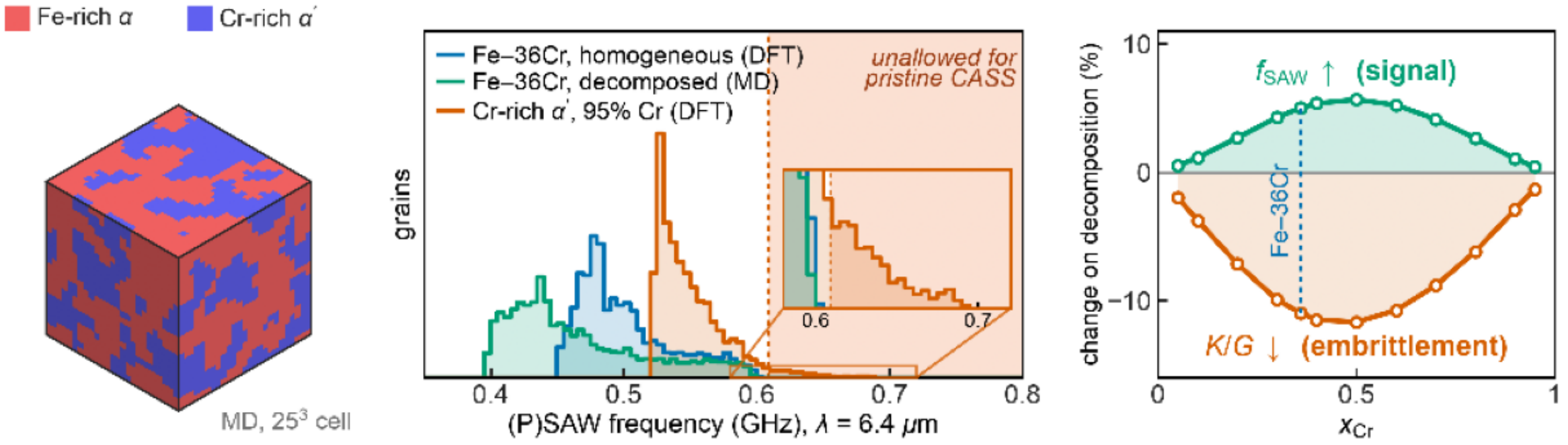
v10



Spinodal decomposition → C' ↑ (stiff α') → f_{SAW} ↑ (signal) + K/G ↓ (embrittlement) → non-destructive detection

One spinodal band: the 748 K CALPHAD spinodal (30-71% Cr, Sec. 3.1) instead of the two 298 K PWSCF-PAW windows of Fig. 1b, whose narrow second window near 95% Cr was confusing; +5.7% under the peak

v11



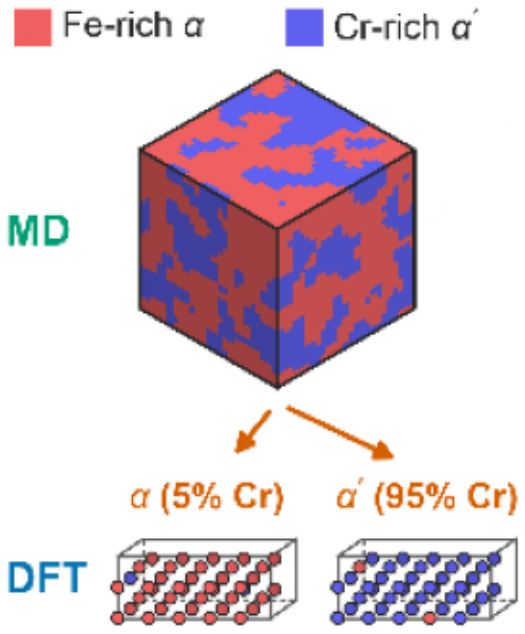
Simulated spinodal decomposition of Fe–36Cr

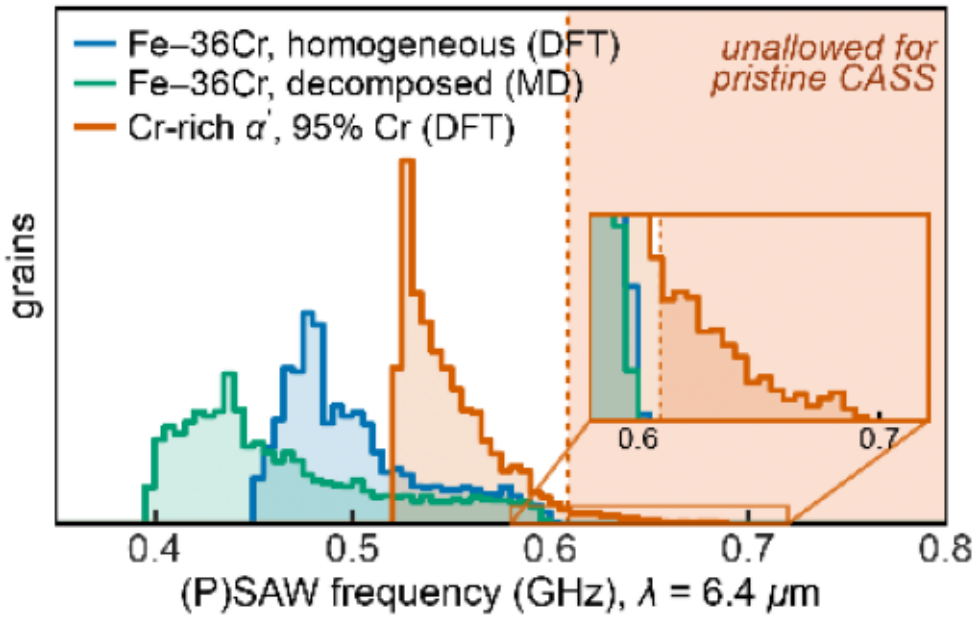
DFT: stiff Cr-rich α′ places surface waves in the range unallowed for pristine ferrite

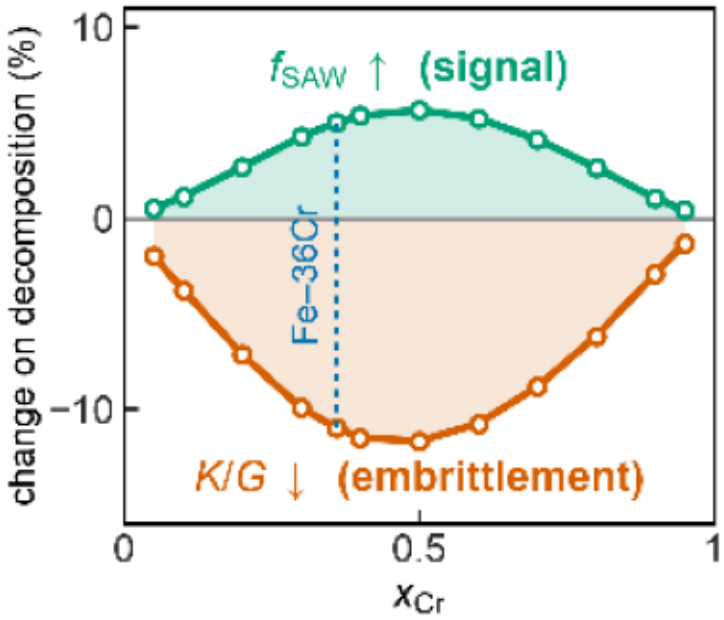
Shear stiffening raises f_{SAW} and lowers K/G: a signature of embrittlement

New design after the Acta 2023 graphical abstract: three panels with bold claims beneath (MD decomposed Fe-36Cr cell; (P)SAW spectrum over grain orientations with inset on the unallowed range, DFT alpha' reaching 0.69 GHz; f_SAW up and K/G down at all twelve MD compositions); no spinodal shading

v12







Simulated spinodal decomposition of Fe–36Cr

DFT: stiff Cr-rich α′ places surface waves in the range unallowed for pristine ferrite

Shear stiffening raises f_{SAW} and lowers K/G: a signature of embrittlement

v11 plus the QE 40-atom cells of alpha (5% Cr) and alpha' (95% Cr) under the MD cube in panel 1, with arrows from the cube (MD separates, DFT models the two phases)